

Nirush Man Shrestha

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GitHub: github.com/ByteNirush | Portfolio: nirushmanshrestha.com.np

Technical Skills

Languages: TypeScript, Java, Python, Go, C, JavaScript, C#

Frameworks: Node.js, Express, Spring, Swing, JavaFX, Django, FastAPI, Flask, React, Tailwind CSS

Databases: PostgreSQL, MySQL, SQLite, Firebase

Cloud & DevOps: AWS (EC2, S3, RDS), Docker, Git/GitHub

Core CS: Object-Oriented Programming, Data Structures & Algorithms, System Design

Soft Skills: Problem Solving, Team Collaboration, Communication, Time Management, Leadership, Critical Thinking

Projects

CollabBoard – Real-Time Collaborative Whiteboard | [GitHub](#)

Tech Stack: Go (chi, Gorilla WebSocket), PostgreSQL (pgx), React, TypeScript, Vite, JWT Auth, Docker

- Engineered a real-time collaborative whiteboard (Go + WebSockets + React/TypeScript) with an event-driven hub architecture, minimizing broadcast fan-out and cutting average message propagation to sub 100ms under multi-user load.
- Optimized backend efficiency by consolidating board/stroke queries, adding indexed migrations, and refactoring hot paths reducing DB round trips and lowering server CPU utilization while keeping codebase modular and testable.
- Implemented secure JWT auth, salted password hashing, and Dockerized deployment pipeline (Go + Postgres + Vite frontend), streamlining local-to-production parity and enabling rapid iterative releases with minimal overhead.

Electricity Billing System | [GitHub](#)

Tech Stack: Java, JavaFX, MySQL

- Designed and implemented relational schema to manage customer data, billing, and tax calculation.
- Built a complete backend that automates billing operations, implements CRUD with SQL queries, and handles real-time data updates. Integrated data validation and reports with SQL queries for performance tracking.
- Demonstrates database design, data manipulation, and backend automation.

Taxi Booking System | [GitHub](#)

Tech Stack: Python, Tkinter, SQLite

- Designed and implemented a desktop-based Taxi Booking System using Python, Tkinter, and SQLite, delivering a reliable, multi-role platform for customers, drivers, and administrators.
 - Applied OOP principles + UML design, improving maintainability and scalability.
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Education

BSc (Hons) Software Engineering

University of Bedfordshire (PCPS, Nepal)

Sep 2024 – Expected 2027

Relevant Coursework: Object-Oriented Programming, Data Structures & Algorithms, Databases, System Design

Certifications

- Career Essentials in Software Development – Microsoft & LinkedIn (2025)
 - Python Essentials 1 – Cisco Networking Academy (2025)
 - AWS Academy Graduate: Cloud Architecting – AWS Academy (2025)
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Achievements

- 2nd Runner-Up, Self-Care Hackathon 2025 – Built and deployed a full-stack Django app in 48 hours.